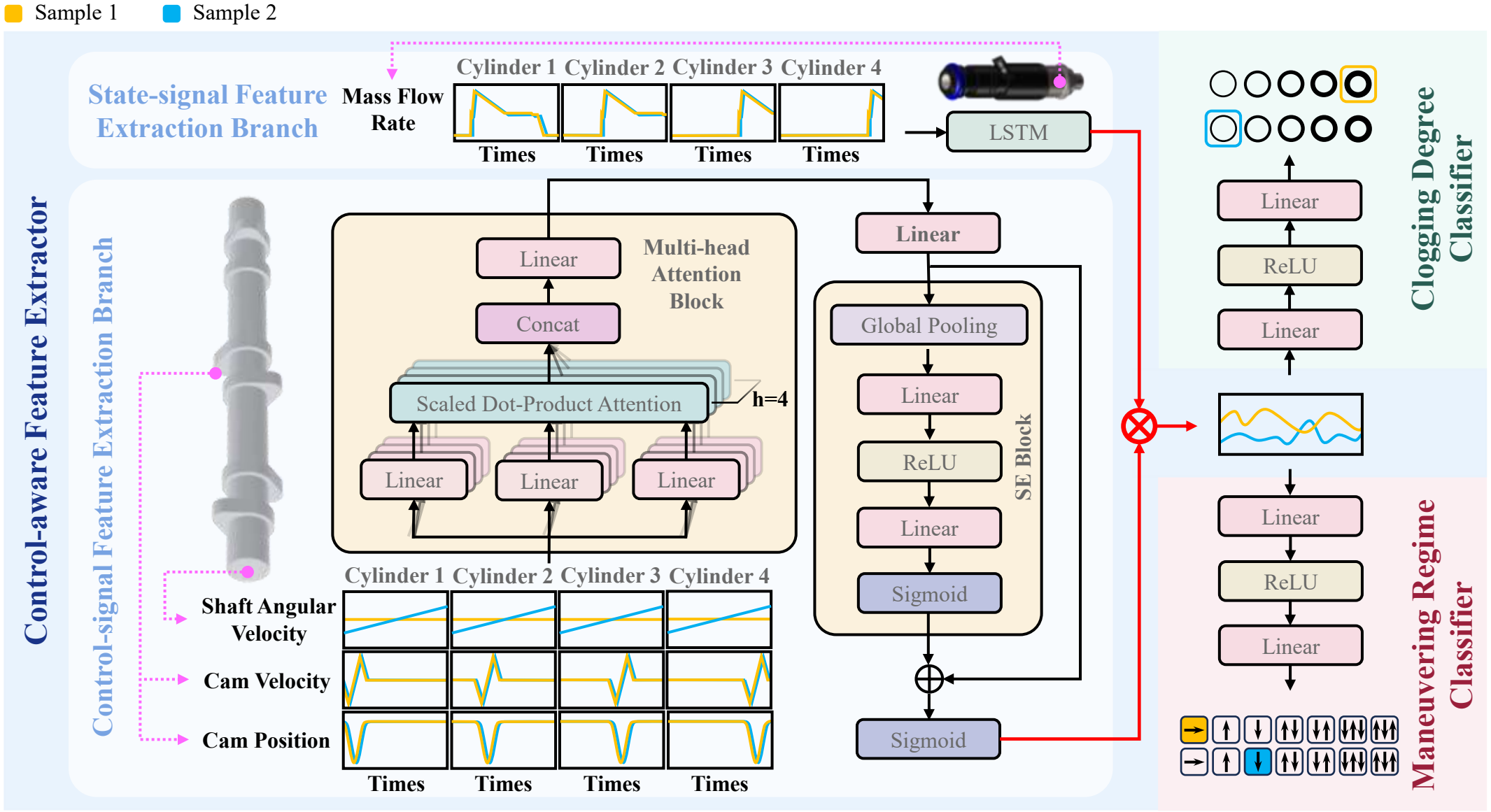

Sample 1
Sample 2
Control-aware Feature Extractor
State-signal Feature Extraction Branch
Mass Flow Rate
Cylinder 1 Cylinder 2 Cylinder 3 Cylinder 4
Times Times Times Times
LSTM
Control-signal Feature Extraction Branch
Linear
Concat
Multi-head Attention Block
Scaled Dot-Product Attention
h=4
Linear Linear Linear
Shaft Angular Velocity
Cam Velocity
Cam Position
Cylinder 1 Cylinder 2 Cylinder 3 Cylinder 4
Times Times Times Times
Linear
Global Pooling
Linear
ReLU
Linear
Sigmoid
SE Block
Sigmoid
Linear
ReLU
Linear
Clogging Degree Classifier
Linear
ReLU
Linear
Maneuvering Regime Classifier